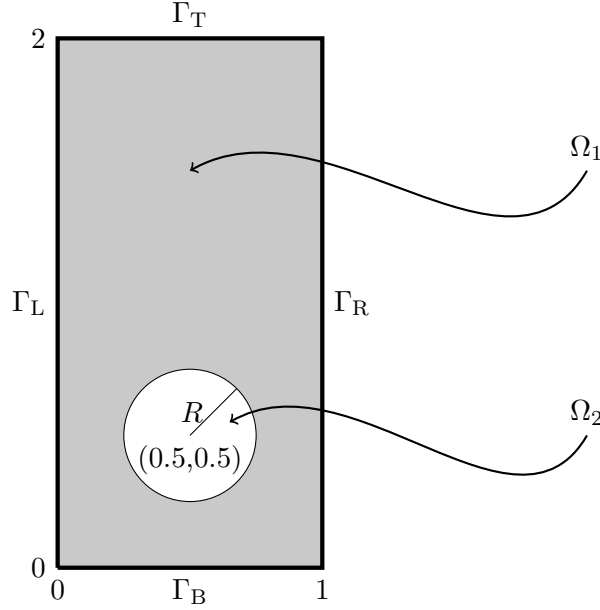


# CFD 2025-26 – Lab 12

Consider a closed box  $\Omega = (0, 1) \times (0, 2)$  in which two immiscible fluids are present, with different densities  $\rho_1$  and  $\rho_2$ , but the same dynamic viscosity  $\mu$ . In the initial condition, the lighter fluid (with density  $\rho_2 < \rho_1$ ) is confined in a circular region  $\Omega_2$ , while the heavier one occupies the remaining part  $\Omega_1$  of the box, as shown in the figure below.



This system can be described by classic Navier-Stokes equations with a variable density  $\rho(t, \mathbf{x})$ :

$$\begin{cases} \rho \frac{\partial \mathbf{u}}{\partial t} + \rho(\mathbf{u} \cdot \nabla) \mathbf{u} - \operatorname{div} (2\mu \varepsilon(\mathbf{u})) + \nabla p = \rho \mathbf{g}, & \text{in } \Omega, \text{ for } t > 0, \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, \text{ for } t > 0, \\ \mathbf{u} = \mathbf{0}, & \text{on } \Gamma_B \cup \Gamma_T, \text{ for } t > 0, \\ \mathbf{u} \cdot \mathbf{n} = 0, \quad \mathbf{n} \cdot 2\mu \varepsilon(\mathbf{u}) \cdot \boldsymbol{\tau} = 0, & \text{on } \Gamma_L \cup \Gamma_R, \text{ for } t > 0, \\ \mathbf{u} = \mathbf{0}, & \text{in } \Omega, \text{ at } t = 0, \end{cases} \quad (1)$$

where  $\varepsilon(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T)$  and the initial density distribution is

$$\rho(0, \mathbf{x}) = \begin{cases} \rho_1, & \text{for } \mathbf{x} \in \Omega_1, \\ \rho_2, & \text{for } \mathbf{x} \in \Omega_2. \end{cases}$$

Due to the action of the gravitational field  $\rho \mathbf{g}$ , the bubble of fluid 2 is going to move in the domain, and the description of this motion has to be accounted for in the fluid equations (1). One way to model this is the *level-set method*, in which the interface  $\partial\Omega_2$  between the two fluids is represented by the zero-level set of a function  $\varphi(t, \mathbf{x})$  that fulfills the following evolution equation:

$$\begin{cases} \frac{\partial \varphi}{\partial t} + \mathbf{u} \cdot \nabla \varphi = 0, & \text{in } \Omega, \text{ for } t > 0, \\ \varphi = \frac{1}{2} [(x - 0.5)^2 + (y - 0.5)^2 - R^2], & \text{in } \Omega, \text{ at } t = 0. \end{cases} \quad (2)$$

This function is a signed distance function from  $\partial\Omega_2$  such that

$$\varphi > 0 \text{ in } \Omega_1, \quad \varphi < 0 \text{ in } \Omega_2, \quad \|\nabla \varphi|_{\partial\Omega_2}\| = 1,$$

therefore we can define the piecewise constant density function  $\rho(t, \mathbf{x})$  at each time  $t$  as

$$\rho(t, \mathbf{x}) = \begin{cases} \rho_1, & \text{if } \varphi(t, \mathbf{x}) > 0, \\ \rho_2, & \text{otherwise.} \end{cases}$$

## Exercise 1

- 1 Derive the weak formulation of the fluid problem (1) and of the level-set problem (2), separately. For simplicity, consider homogeneous Neumann conditions on the whole boundary for (2).
- 2 Introduce a finite element space discretization for both problems.
- 3 Consider the coupled system made of both semi-discrete formulations and discretize it in time with the backward Euler scheme, with a treatment of the advection term that is:
  - semi-implicit for the fluid problem;
  - implicit for the level-set problem.
- 4 Repeat the previous points with different coupling strategies, that is, with different kinds of treatment of the advection terms. What are the pros and cons of each choice?
- 5 Compute the total mass  $m = \int_{\Omega} \rho \, d\Omega$  of the system every tenth time step. What can you observe?